



PAWMAP: A Community-Driven Stray Animal Rescue and Welfare Platform

By

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Under the guidance of

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A Project report submitted in partial fulfillment of the requirements for
the award of degree of Master of Science (Data Science) of CHRIST
(Deemed to be University)

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CERTIFICATE

*This is to certify that the report titled **PawMap** is a bona fide record of work done by **Anavadhya Devi (2548007)**, **Sreya Anil (2548052)** of **CHRIST(Deemed to be University)**, Bengaluru, in partial fulfillment of the requirements of IV Trimester M.Sc. (Data Science) during the year 2026-27.*

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Abstract

India hosts one of the largest stray animal populations in the world, with tens of millions of stray dogs and cats across its cities. Despite active NGO and volunteer efforts, rescue coordination currently relies on fragmented channels such as WhatsApp groups and phone calls, resulting in duplicated responses, missed cases, lost medical histories, low adoption rates, and an absence of data to guide intervention. PawMap is proposed as a community-driven mobile and web platform designed to address these gaps by unifying stray animal reporting, volunteer dispatch, medical record-keeping, lost-and-found matching, and adoption facilitation within a single system. Built on Django REST Framework, React/React Native, PostgreSQL with PostGIS, and supporting services such as Celery, Redis, and Google Maps API, the platform enables geo-tagged case reporting in under a minute, atomic case-claiming with automatic reassignment, caregiver-independent medical records, a weighted temperament safety rating, and a foster-to-adopt pipeline. A data science layer applies DBSCAN spatial clustering to detect abandonment hotspots, performs coverage gap analysis, and computes volunteer reliability scores to prioritise case assignment. Initially scoped for pilot deployment in Bengaluru, PawMap is designed for future scaling to other Indian cities, offering NGOs an evidence-based tool for coordinated, data-driven animal welfare intervention.